

FastTrack

Phase 3

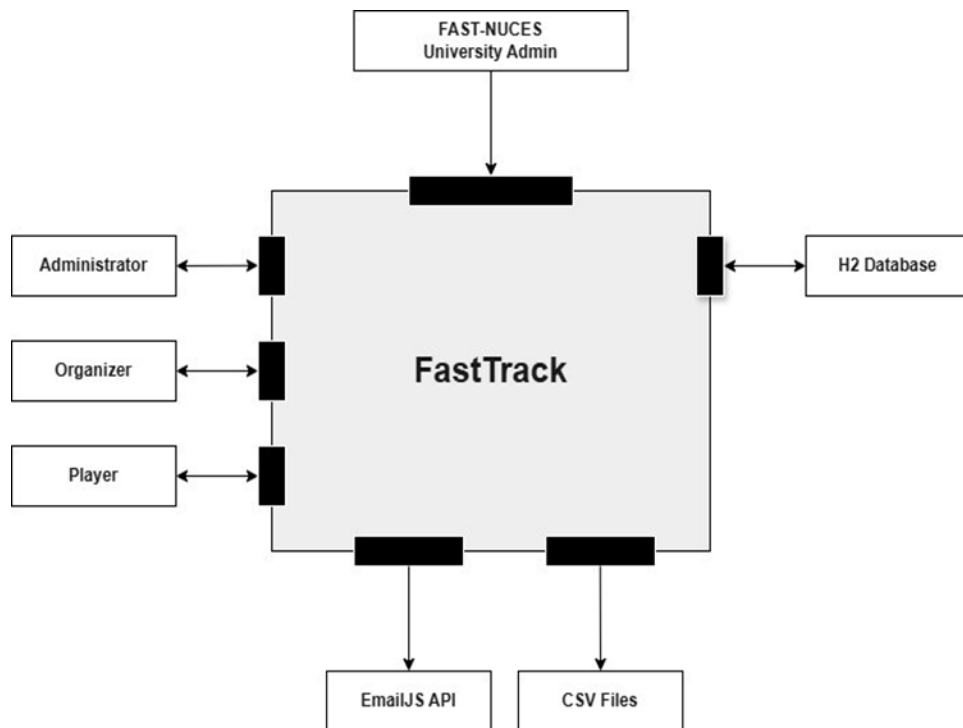
Team 3

Member Name	Member Roll #	Primary Responsibility
Syed Saad Ali	24L-2549	ArCD,B&A,Description
Abdul Ahad	24L-3029	ArCD,B&A,Description
Hassan Ashraf	24L-3073	ArCD,B&A,Description
Romesha Afzal	24L-3001	ArCD,B&A,Description
Kabeer Ahmed	24L-3087	Rationale

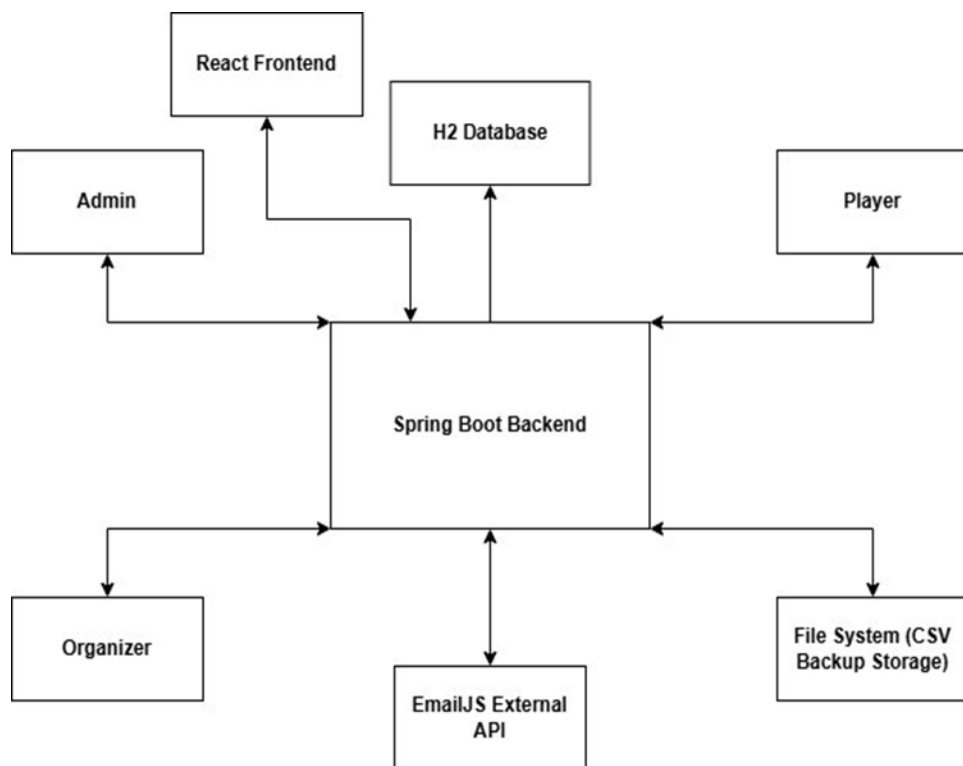
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1. Architecture Context Diagram (ArCD)



2. Box and Arrow Diagram



3. Description

Sr#	Component Name	Component Description	Requirement Reference
1	User Authentication & Account	Handles user registration (Player/Organizer), login, account deletion, profile management, and role-based access control.	UC-01, UC-02, UC-03, UC-14, UC-15, UC-18 NFRs: 14, 15, 17, 30, 31 (Security, Usability)
2	Profile Management	Allows users (Admin, Organizer, Player) to view and update personal/academic details, profile picture, skills, and preferences.	UC-03, UC-05 NFRs: 10, 11, 18 (Usability, Security, Privacy)
3	Search & Browse	Provides global search and browsing functionality for events, teams, players, and profiles with filters.	UC-04, UC-06, UC-07 NFRs: 4 (Performance)
4	Event Management	Enables creation, viewing, updating, and management of sports events (details, schedule, registration status, etc.).	UC-06, UC-07, UC-19 to UC-24 NFRs: 1, 3, 5, 28 (Performance, Responsiveness)
5	Team Management	Supports team creation, member management, role assignment, join requests, captain functions, and team details.	UC-16, UC-24 to UC-28, UC-30, UC-31 NFRs: 29 (Data Integrity)
6	Registration & Participation	Manages individual/team event registrations, cancellations, and registration portal.	UC-14, UC-15, UC-26, UC-33, UC-36 NFRs: 9, 12 (Usability)
7	Match & Schedule Management	Handles match scheduling, recording results, tournament types (Knockout, Round Robin), and venue assignment.	UC-09, UC-11, UC-12, UC-32, UC-37, UC-38, UC-39 NFRs: 2, 3, 7 (Performance)
8	Notification System	Sends and manages system notifications, event updates, registration confirmations, and custom broadcasts.	UC-08, UC-13 NFRs: 5 (Responsiveness)
9	Leaderboard & Results	Displays rankings, match results, and tournament standings.	UC-09, UC-10 NFRs: 7 (Performance)
10	Admin & Organizer Dashboard	Provides administrative functions like managing users, organizers, events, reports, and system oversight.	UC-17 to UC-23, UC-33 to UC-39 NFRs: 15, 19, 27 (Security, Scalability)

4. Rationale

FastTrack is a web application that follows the Client-Server architectural pattern. The Spring Boot backend acts as the central server holding all resources - user data, events, teams, matches, and notifications all persisted in the H2 database. The React frontend acts as the client, making HTTP requests via Axios to the backend's REST API endpoints whenever a user performs any action such as logging in, registering for an event, or joining a team. The server then processes the request and responds accordingly.

This is consistent with the definition: "a server has resources and a client requests the server for a particular resource, the server processes the request and responds back." FastTrack's three user roles Admin, Organizer, and Player all interact with the system through this same request-response mechanism, making the pattern a natural and accurate fit for the application's architecture.